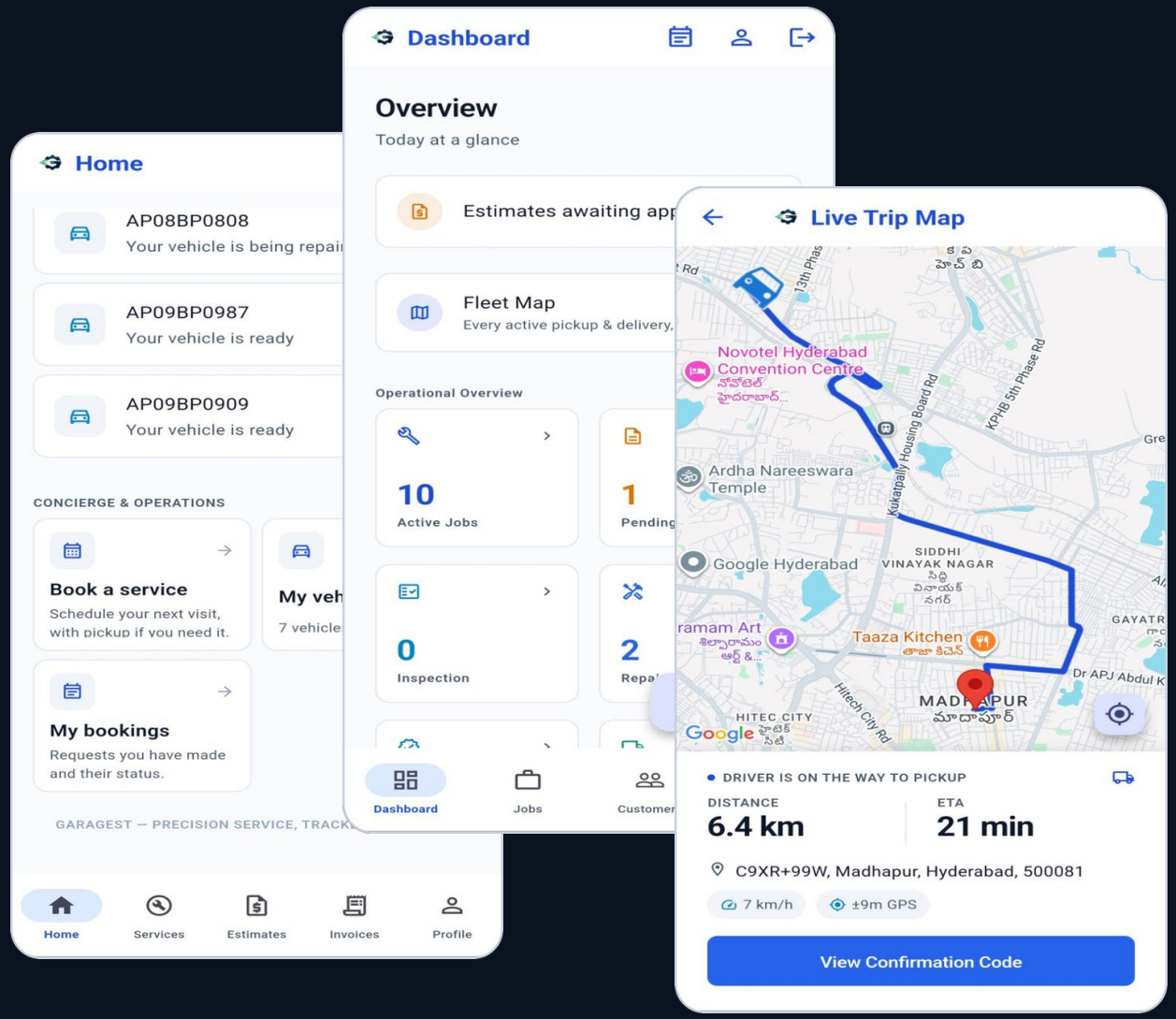
GarageST
SMART GARAGE MANAGEMENT

GarageST

Smart Garage Management Platform

Digitizing the complete vehicle service journey.



TVASTRI

The garage today

Running a workshop on paper, calls and memory makes the same problems repeat.

- **Manual job tracking**
- **Scattered customer communication**
- **Paper and checklist dependency**
- **Delayed approvals**
- **Unclear repair status**
- **Communication gaps**
- **Pickup and delivery coordination**
- **Limited service visibility**

Tracking operational progress across people, vehicles and approvals becomes difficult.

The GarageST vision

Owner

Technician

Manager

Driver

One vehicle
One job card

Service Advisor

Customer

What is GarageST?

A digital operating platform for modern automotive workshops and service centers — from the first booking to the delivered vehicle.

- Customers get visibility and digital approvals
- The workshop gets control of every job
- Technicians get structured tasks
- Drivers get connected pickup and delivery
- Owners get one view across the business

12 journey steps from booking to closure

9 roles, each with a workspace that fits the job

32 workflow events that can notify the right person

End-to-end service journey

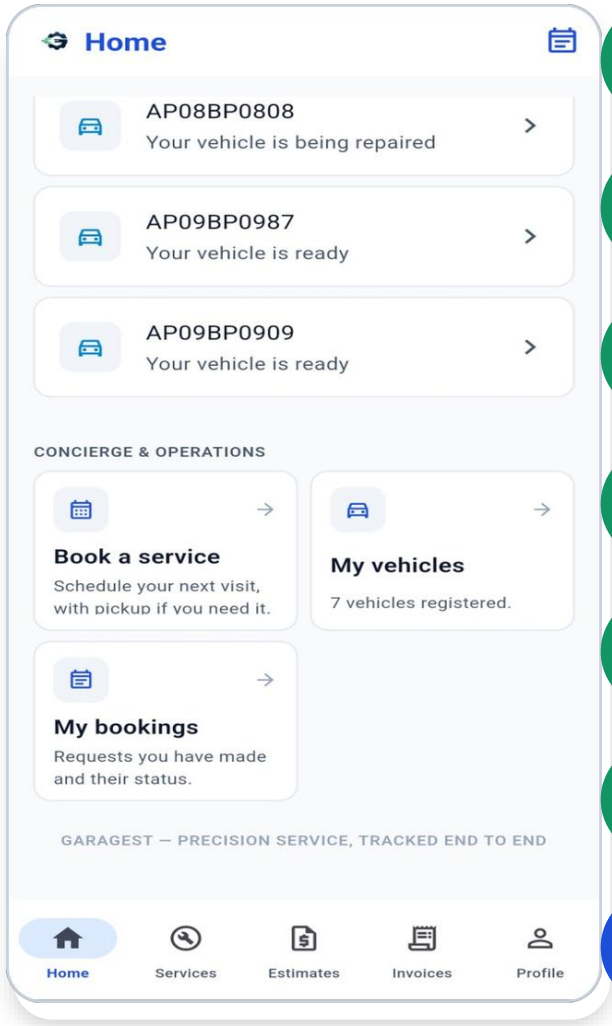
1. Booking
2. Pickup
3. Check-in
4. Inspection
5. Estimate
6. Customer approval
7. Repair
8. Quality check
9. Invoice
10. Payment
11. Delivery
12. Job closure

Pickup applies when the customer asks for it; every step is tracked against the same job.

One vehicle. One digital journey.

- Inspection
- Estimate
- Repair tasks
- Quality check
- **JOB CARD** — the single record for the vehicle's visit
- Invoice & payment
- Photos & videos
- Pickup & delivery trips
- Notifications

The customer experience



Book a service
optionally with vehicle pickup

My vehicles & job cards
service visibility in one place

Repair progress
follow the work on your vehicle as it happens

Review & approve estimates
approve or reject digitally

Invoices & payment status
accept the invoice, track payment

Live vehicle journey
follow pickup or delivery on a map

Notifications
important updates reach the phone

Smart inspection

Vehicle checked in → Inspection checklist → Findings recorded → Photos / videos attached → Inspection completed → Estimate prepared

Structured checklist

Inspection templates guide the technician through each area.

Findings per complaint

Each customer complaint is inspected and recorded.

Evidence that travels with the job

Photos and videos are stored against the job card, with time and location details.

Digital estimate & approval

Garage prepares estimate → Customer reviews items → Approve or reject → Manager confirms → **Repair begins**

For the customer

- Itemised estimate on the phone
- Notified when the estimate is ready
- Decide digitally — no chasing calls

For the workshop

- Managers and the owner are notified of the decision
- A manager confirmation gate before repair starts
- Approval and repair stay clearly separate steps

Repair management

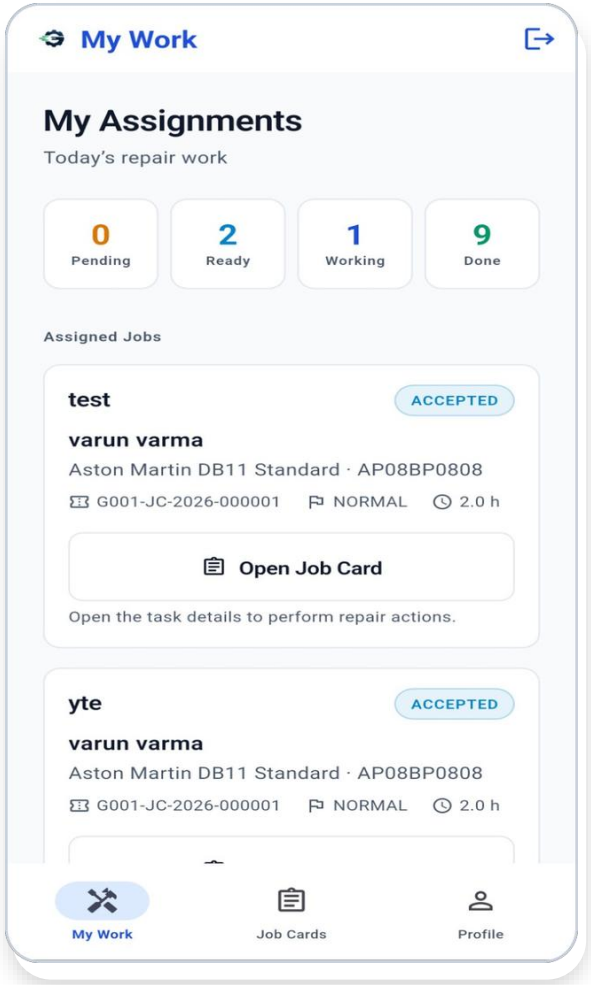
Job card

Repair pending → Repair in progress → Repair completed

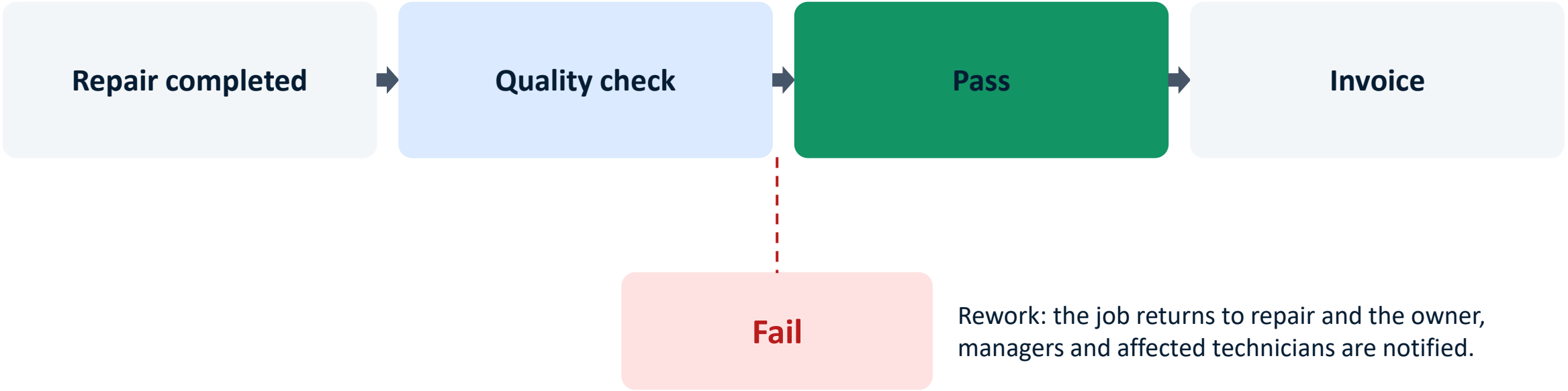
Repair tasks (per technician)

Assigned → Accepted → In progress → Completed

- Tasks are assigned to technicians, who accept, start and complete them
- The job moves to “repair completed” only when every task is complete
- Managers see progress without asking around



Quality control



Rework: the job returns to repair and the owner, managers and affected technicians are notified.

- Quality is verified before the customer is invoiced
- Pass or fail is recorded against the job
- Managers are told when a check starts, passes or fails

Billing & invoice

Invoice generated → Customer views & accepts → Payment recorded → Vehicle ready for delivery

Invoice from the approved estimate

Items match what the customer approved.

Customer visibility

Invoices and payment status are visible in the customer app.

Notified at each step

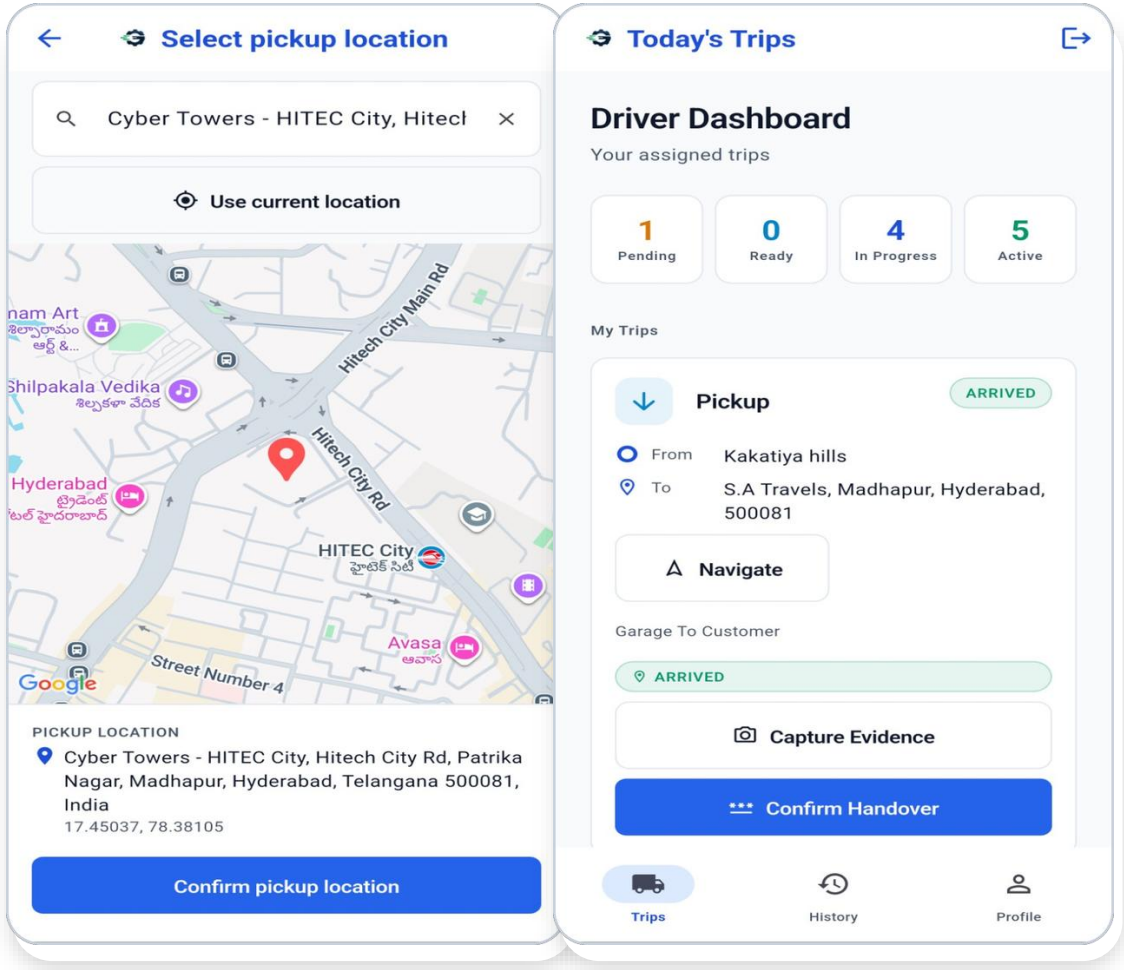
Invoice ready · invoice accepted · payment received.

Payments are recorded inside GarageST, in full or in part, with a payment history. An online payment gateway, refunds and receipt PDFs are not part of the current product (see roadmap).

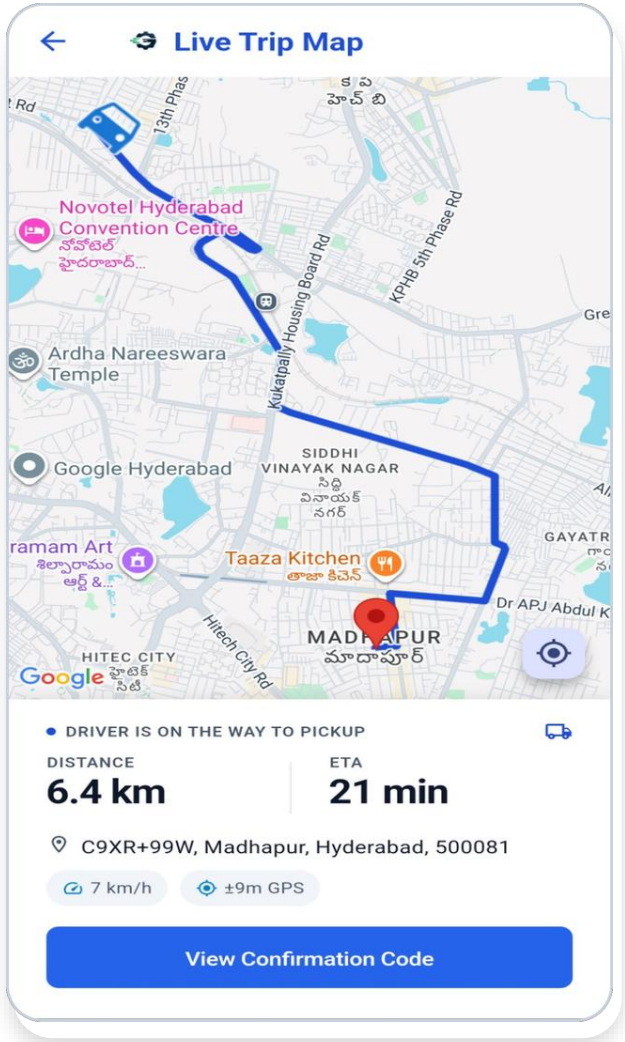
Vehicle pickup

Pickup request → Garage confirms → Driver assigned → Trip in progress → Handover

- Customer chooses the pickup point on a map
- The garage confirms and assigns a driver
- The driver accepts and starts the trip
- Customer and workshop are notified at each stage



Live vehicle tracking



Live driver location

the vehicle moves on the map as the driver drives

Route, distance and ETA

shown to the customer on the trip screen

Customer visibility

the customer follows their own vehicle’s trip

Workshop visibility

managers and owners see active trips on a fleet map

Available during active pickup and delivery trips.

Digital handover

Driver arrives

the customer is notified

Evidence photos

condition captured at handover

Customer shows a code

generated in the customer’s app

Driver verifies

handover confirmed in the system

A handover verification code confirms that the vehicle really changed hands — for both pickup and delivery. It is a vehicle-handover check, not a login code.

Delivery management

Ready for delivery → Delivery requested → Driver trip → Handover verified → **Vehicle delivered** → Job closed

- Once payment is recorded the vehicle is marked ready for delivery and the customer is notified
- Delivery uses the same driver, live-trip and handover tools as pickup
- Completing the verified delivery trip records the delivery and marks the job delivered
- The job is then closed by the workshop



NEW

Real-time notifications

GarageST proactively notifies users about important workflow events — on the phone's notification tray, even when the app is not in the foreground.

Bookings

requested · confirmed · rejected · cancelled

Pickup & trips

driver assigned · started · arrived · handover

Inspection

started · completed

Estimates

ready · approved · rejected

Repair

task assigned · started · completed

Quality check

started · passed · failed

Invoice & payment

generated · accepted · payment received

Delivery & closure

ready · delivered · job closed

Delivered through Firebase Cloud Messaging to the user's Android phone.

Notifications by persona

Customer

service updates · estimate ready · invoice · payment · vehicle ready · delivery

Owner

important business events: new booking, estimate decisions, repair done, quality-check failure, invoice, payment, delivery, closure

Manager / Service Advisor

operational updates across bookings, trips, inspection, estimates, repair tasks, QC, invoicing

Technician

new repair task assigned · quality-check rework

Driver

pickup assigned · customer handover ready

Relevant, not noisy

Each person receives only the events relevant to their role and their own garage. You never hear about your own action.

ILLUSTRATIVE

The notification experience

GarageST · now

Your vehicle is ready for delivery

Your vehicle is ready (JC-…).

tap

GarageST opens on the right screen for your role

- **Sound**
- **Vibration per device mode**
- **System tray**
- **Foreground & background**
- **History & unread badge**
- **Deep-link navigation**

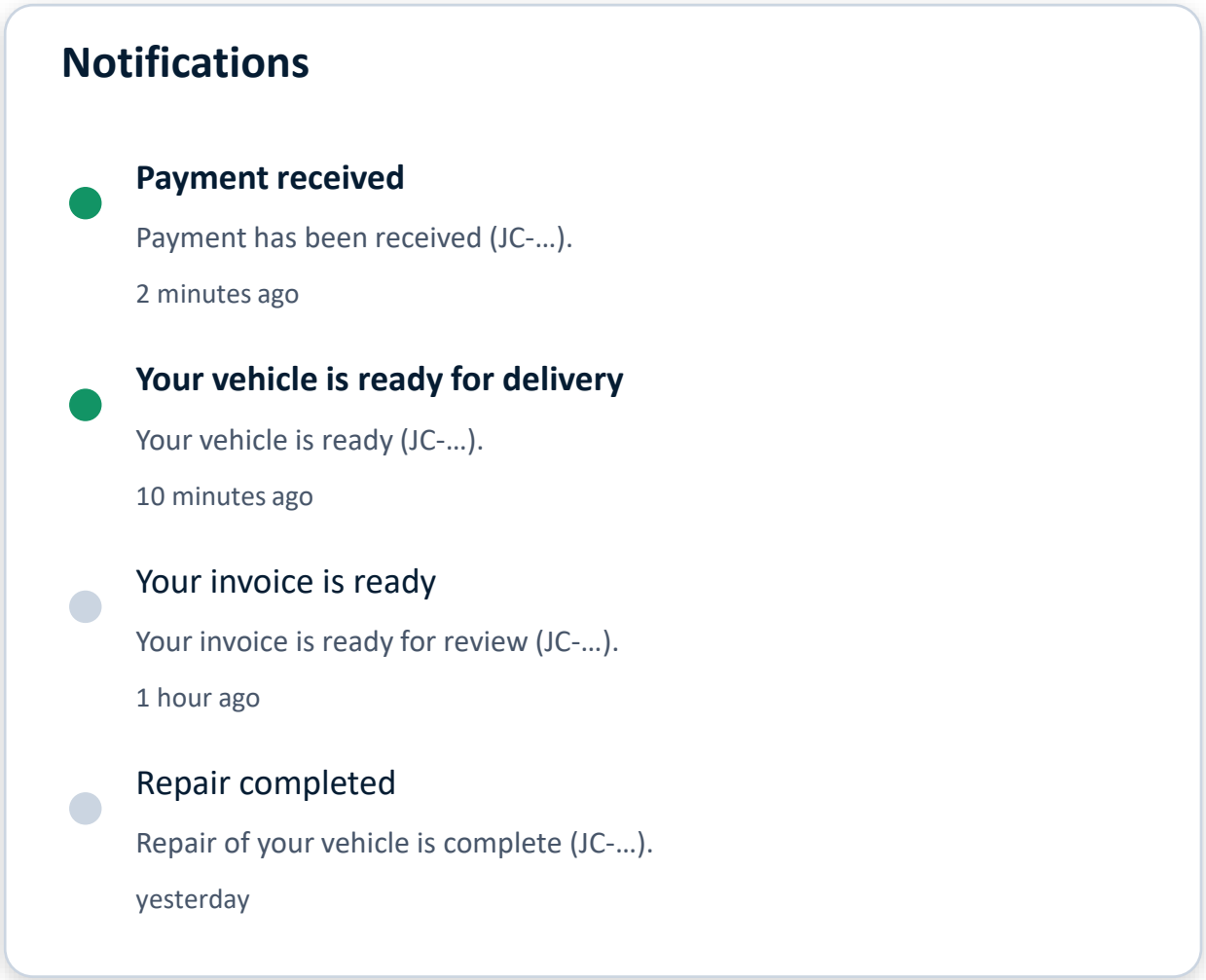
Validated on a real Android device

- Notification appears with sound
- Vibrate / silent follows the phone's mode
- Shown in the system tray
- Tap opens GarageST and the right screen
- History and unread badge work
- Vehicle ready & payment received observed

Illustration drawn for this presentation; wording is the real notification text.

Sound and vibration remain under the user's Android notification settings.

Notification history



Unread indicator

new items stand out; the bell shows the unread count

Mark as read

tap an item to open it and mark it read

Mark all read

clear the list in one tap

Always available

history stays in GarageST even if a push was missed

Illustration — drawn for this presentation using real notification wording.

Communication without chasing

Without it

- Calling the customer to approve an estimate
- Asking the technician if the job is done
- Telling the customer the vehicle is ready
- Checking whether payment came in

With GarageST

- The estimate-ready notification reaches the customer’s phone
- Managers are told when repair or QC changes
- The customer hears “vehicle ready” as soon as it is
- Owners see payments as they happen

Important workflow events can proactively reach the right user instead of depending entirely on manual calls and messages. Today this is in-app push; other channels are not part of the current product.

Media & service evidence

Photos & videos from the job

Inspection, before/during/after-service and trip evidence.

Time and location details

Evidence carries capture time and location information.

Reliable uploads

Media uploads in the background and retries if the network drops.

Private storage

Media is kept in a private bucket and opened through short-lived secure links.

Customer can view

Customers can see the media published for their own job.

Vehicle trip evidence

Pickup and delivery handover photos are kept with the trip.

Service report and branded invoice

Final service report

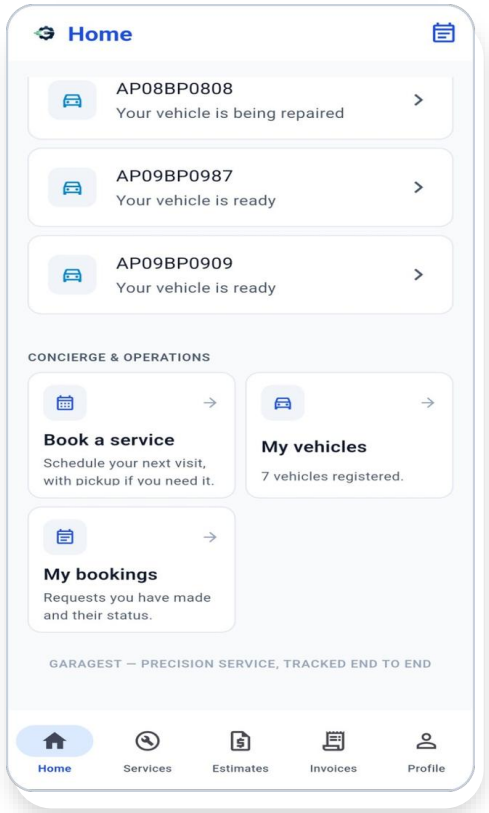
- Created automatically when a job is closed
- A customer version and a garage (owner) version
- The customer version leaves out internal notes and internal photos
- Delivered as a PDF; e-mailed when e-mail is enabled for the garage

Branded invoice

- Garage name and logo up front
- Clear itemised amounts with GST
- Payment status shown on the invoice
- Page numbers and a tidy one-page layout for normal invoices

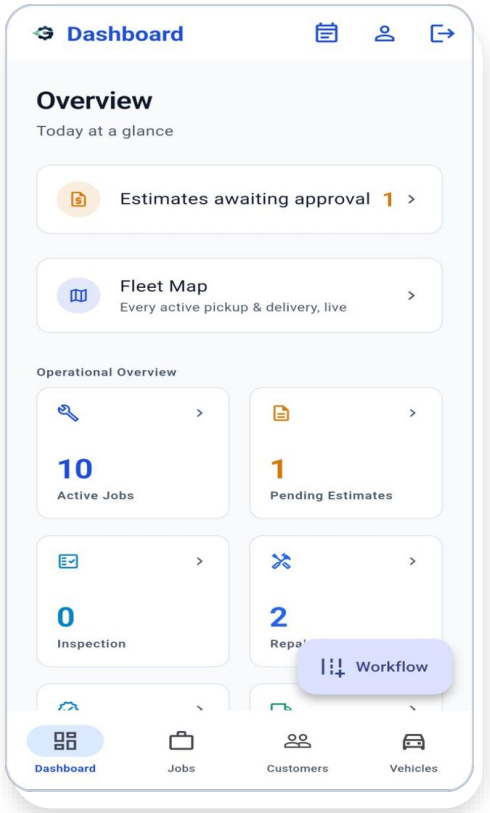
Documents are produced from the same records the team works with — nothing is typed twice.

A workspace for every role



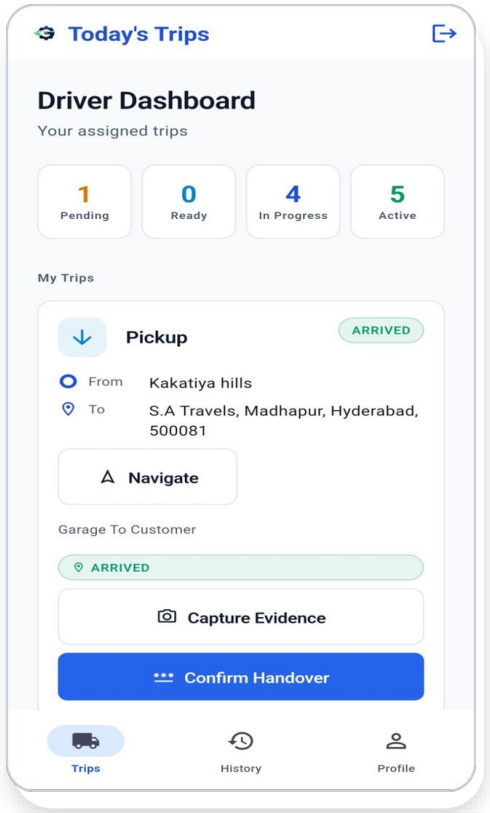
Customer

bookings, vehicles, progress



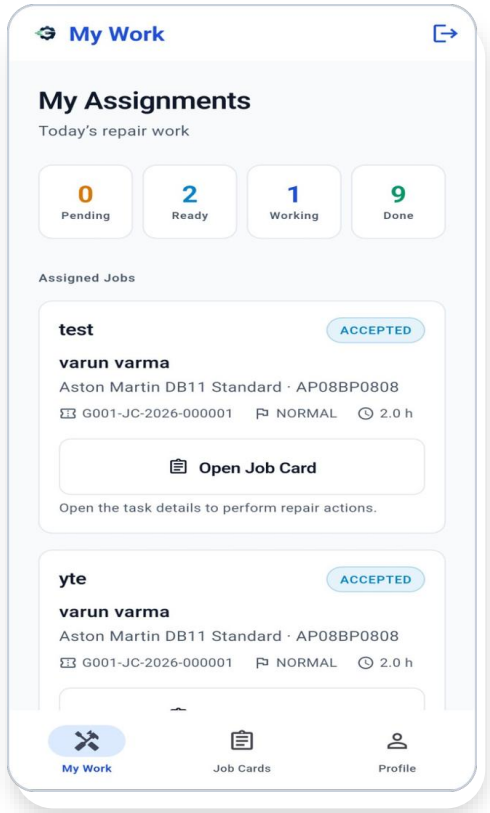
Manager / Owner

jobs, estimates, fleet



Driver

assigned trips



Technician

assigned repair work

Finance and inventory workspaces

Cashier

- Invoices ready for payment
- Full or part payments by cash, UPI, card or bank transfer
- Protection against double payments
- Today's collection and payment history

Accountant

- Review payments and transactions
- Outstanding amounts at a glance
- Payment-method summary
- GST and invoice summary by period
- View-only: reviews, never collects

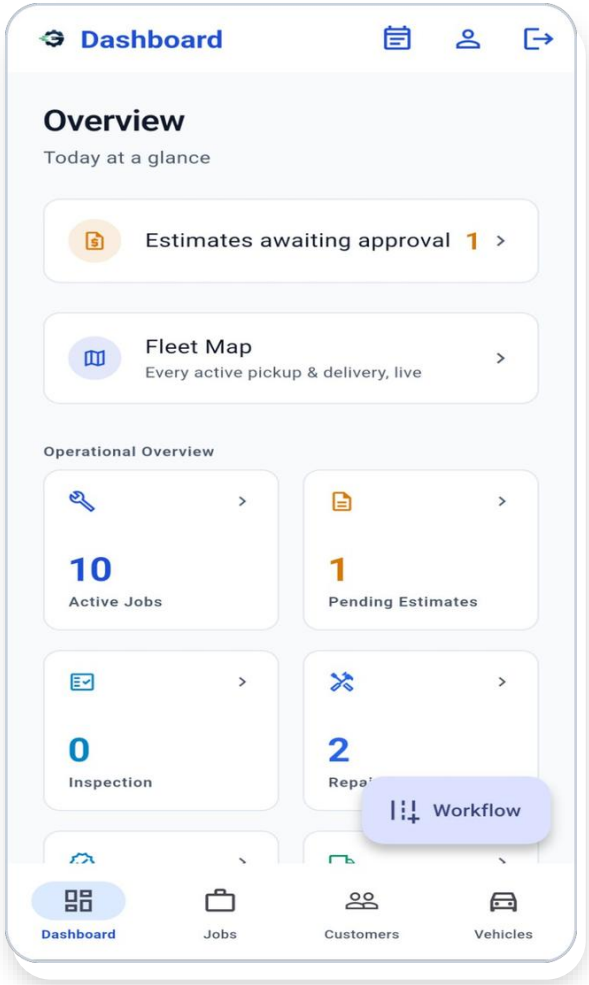
Inventory manager

- Parts catalogue with stock levels
- Stock in, stock out and adjustments
- Movement history
- Low-stock and out-of-stock lists
- Issue parts against a job card

Each role gets its own workspace. What each person can actually do is enforced by the platform.

Benefits for the garage owner

- Centralized visibility of every job
- Job status and workflow tracking
- Staff coordination across roles
- Customer transparency builds trust
- Billing and payment visibility
- Vehicle movement visibility
- Notifications for important business events



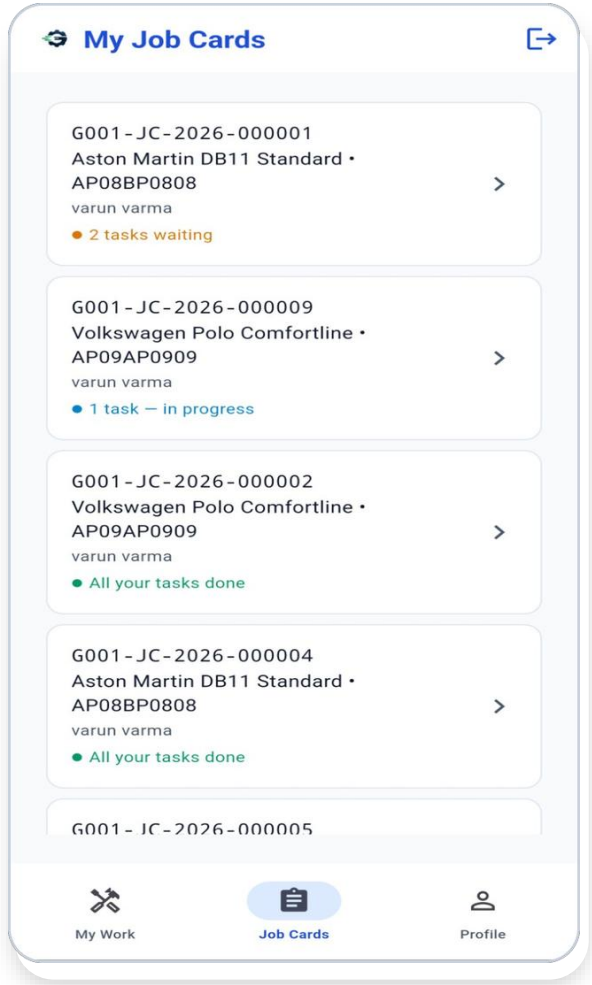
Benefits for managers & service advisors



- Job progress at a glance
- Customer approvals and estimate status
- Repair progress and technician workload
- Quality-check results
- Billing and delivery coordination
- Notified when something needs attention

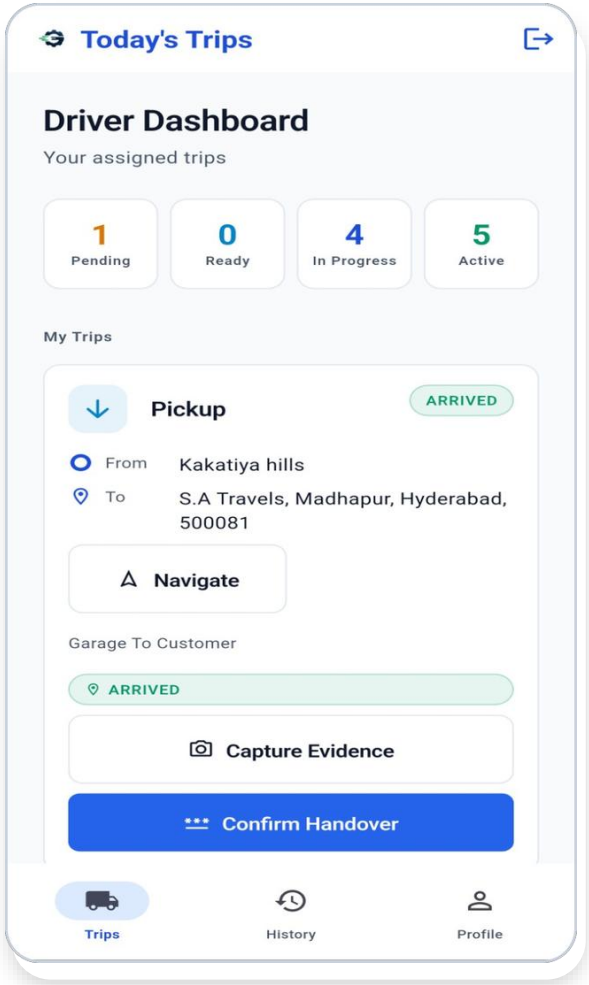
Benefits for technicians

- See the repair tasks assigned to you
- Task status: accept, start, complete
- Job and vehicle context in one place
- Clear workflow progression
- Quality-check feedback when rework is needed
- Notified when a task is assigned



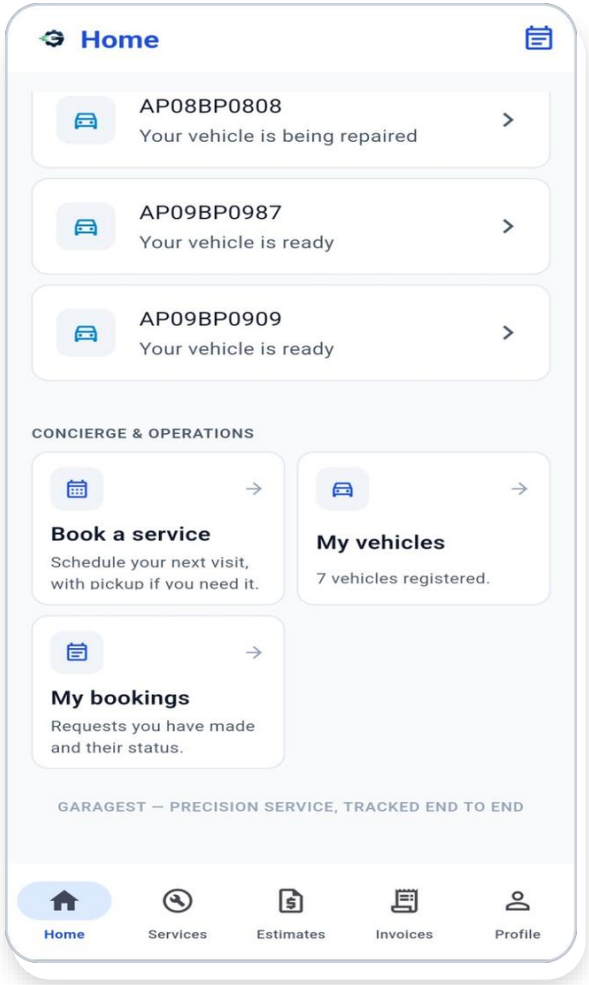
Benefits for drivers

- Assigned pickup and delivery trips
- Navigation to the customer or garage
- Live trip with route
- Evidence photos and verified handover
- Pickup and delivery in one workflow
- Notified of new assignments



Benefits for customers

- Clear visibility of your vehicle’s service
- Digital estimate review and approval
- Service progress without calling
- Invoice and payment status
- Follow your vehicle’s pickup or delivery
- Notifications at the important moments
- Your job-card history in the app



One customer, more than one garage

My Garages

- The garages that already look after your vehicles
- Your favourite is marked as the default
- Booking starts from your default garage

Find another garage

- Your choice, never pushed on you
- Active garages near you, nearest first, with real distances
- About 40 km by default
- No location? Choose one on the map

You stay in control

- Browsing never adds a garage to your list
- A garage joins when it registers you, services your vehicle or confirms a booking
- Your history stays with you across garages

Multi-garage operations

Garage A

- Its own staff and roles
- Its own jobs and trips
- Its own notifications

Garage B

- Its own staff and roles
- Its own jobs and trips
- Its own notifications



Owner

An owner can view the garages they belong to.

- Staff see only their own garage’s work
- A notification for Garage A never reaches Garage B staff
- Customers see only their own vehicles and jobs

Advanced branch / franchise management features are not part of the current product.

TRUST

Security & data control

Authenticated access

Every user signs in; sessions use secure tokens.

Role-based permissions

People see and do what their role allows — enforced by the platform, not just the screen.

Garage-level isolation

Customers, vehicles, staff, estimates, deliveries and payments stay within the garage.

Customer ownership

Customers only reach their own vehicles and jobs, even across several garages.

Secure communication

The app talks to the platform over secured APIs.

Controlled media access

Photos and videos are private and opened with short-lived links.

Notification ownership

You can only read and manage your own notifications.

No secrets in the app

Service credentials stay on the server.

Verified sign-up and secure handover

SMS verification

- A one-time code confirms the mobile number at registration
- Codes expire quickly and are limited per number
- Available when switched on for the deployment

Handover code

- The customer shows a code at pickup and delivery
- The driver confirms it before the vehicle moves on
- Photos record the vehicle’s condition

Private by design

- Customers only reach their own vehicles and jobs
- Garages only reach their own customers and vehicles
- Owners see the garages they belong to

Technology foundation

Flutter

mobile app

Spring Boot

application platform

PostgreSQL

system of record

Firebase Cloud Messaging

push notifications

Cloudflare R2

media storage

Render

hosting

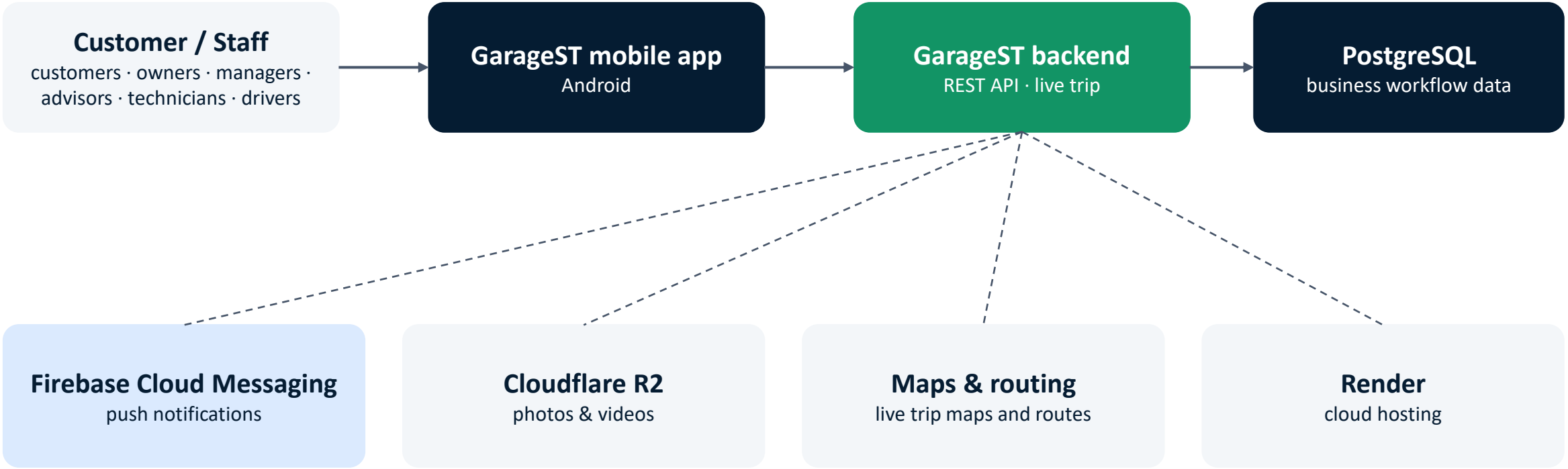
STOMP / WebSocket

live trip tracking

Maps & routing services

maps, routes, geocoding

How everything connects



The backend owns the business rules; PostgreSQL is the source of truth; notifications, media, maps and hosting are supporting services.

Current product status

Available today

- Booking, job-card workflow and role-based workspaces
- Digital inspection, estimate and customer approval
- Repair tasks, technician assignment, quality check
- Invoicing, payments (full or partial) and branded invoice PDF
- Cashier, accountant and inventory-manager workspaces
- Customer garages: My Garages, default garage, nearby discovery
- Pickup and delivery trips, live tracking, verified handover
- Photos and videos as service evidence; final service reports
- Push notifications and in-app history (validated on Android)
- Garage-level isolation; owners can view their garages

Not part of today’s product

- Online payment gateway
- Refunds, voids and receipt PDFs
- Supplier and purchase-order management
- iOS push notifications
- WhatsApp / e-mail verification codes
- Notification preferences and quiet hours
- Advanced branch / franchise management

Listed as roadmap candidates on the next slide — not commitments.

Roadmap

Now
In the product today

- Complete service journey
- Role-based workspaces incl. finance and inventory
- Customer garages and nearby discovery
- Live trips and handover
- Push notifications on Android

Next
Candidate next steps

- iOS push notifications
- Online payment gateway
- Receipts and refunds

Future
Longer-term options

- Supplier and purchase management
- WhatsApp / e-mail channels
- Notification preferences

Directions under consideration, each traceable to a known gap in today's product; no dates or commitments are implied.

Why GarageST

- Connected service lifecycle
- Role-based workflow
- Digital inspection
- Approval workflow
- Repair task management
- Live vehicle movement
- Digital handover
- Billing visibility
- Proactive push notifications
- Service evidence
- Customer transparency
- **One connected platform**

Getting a garage onto GarageST

1



Garage setup

register the garage; staff join and are approved

2



User roles

owner, managers, advisors, technicians, drivers, cashiers, accountants, inventory managers

3



Service workflow

the guided job-card workflow

4



Inspection templates

checklists for technicians

5



Customers & vehicles

added by staff or by customers

6



Staff operations

jobs, estimates, repairs, invoices

7



Customer experience

customers register in the app

8



Notifications

enabled for the garage’s environment

9



Go-live

start running real jobs

Measuring success

Job turnaround time

Approval response time

Inspection completion

Repair task completion

Delivery turnaround

Notification delivery & read rate

Customer digital engagement

Operational visibility

These are measures a garage could track with GarageST. They are suggestions, not results achieved by any customer.

GarageST

SMART GARAGE MANAGEMENT

From vehicle arrival to vehicle delivery —

one connected digital journey.

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